

❏ Detailed Theory: Log Analyzers & SIEM

❏ 1. What is a Log?

A **log** is a time-stamped record of events and actions generated by software, systems, applications, or devices. Logs provide insight into system operations, security issues, and user behaviors.

❏ Common Log Types:

Log Type	Description
System Logs	OS-level activities, kernel events, boot logs
Security Logs	Auth attempts, firewall events, IDS alerts
Application Logs	Errors, warnings, and activity from apps
Web Server Logs	HTTP requests, errors, source IP
Audit Logs	Tracks who did what and when

☒ 2. Introduction to Log Analyzers

Log analyzers are tools or systems that parse, filter, and visualize logs for:

- Troubleshooting
- Intrusion detection
- Trend analysis
- Performance monitoring

☒ Common Log Analyzers:

- ELK Stack (Elasticsearch + Logstash + Kibana)

- Graylog
 - Splunk (Free and Enterprise versions)
 - GoAccess (for web logs)
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☒ 3. Introduction to SIEM (Security Information and Event Management)

SIEM is a centralized platform that combines:

- Log Management
- Event Correlation
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Security Monitoring

- Alerting and Reporting

It helps detect suspicious activity by aggregating and analyzing logs from various sources (firewalls, endpoints, network devices, etc.)

☒ Examples of SIEM Solutions:

SIEM	Description
Splunk	Enterprise-level log analysis and real-time correlation
IBM QRadar	Threat detection, behavior analysis
ArcSight	Advanced correlation engine
OSSIM (by AlienVault)	Open-source SIEM
Wazuh + ELK	Free, scalable SIEM stack

❏ 4. Log Forwarding Configuration (in SIEM)

To get logs into a SIEM system, **log forwarding** needs to be configured:

❏ **Methods:**

- Syslog Protocol (UDP/514 or TCP/514)
- Agent-based forwarding (e.g., OSSEC, NXLog, Wazuh agent)
- Log shippers (Logstash, Fluentd, Filebeat)
- APIs or Push Connectors

❏ **Configuration Steps:**

1.
Enable and configure logging on source device (e.g., firewall, server)

2.
Define log forwarding rule (destination IP/port of SIEM)
 3.
Choose log format (CEF, LEEF, JSON, raw)
 4.
Verify delivery using tools like tcpdump or logger
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❏ 5. Log Correlation in SIEM

Log Correlation is the process of combining related events across multiple systems to identify complex attack patterns or misbehavior.

❏ Example:

- Multiple failed logins + privilege escalation + outbound connection = Possible intrusion

❏ How It Works:

- **Correlation Rules** are defined using logical conditions
 - **Normalized data** is used to create relationships
 - **Time windows** help identify sequences of events
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☒ 6. Event Triggering in SIEM

When correlated patterns match certain criteria, **event triggers** are fired. These lead to:

- **Alert generation**
- **Incident tickets**
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Automated responses (SOAR)

- Dashboard updates

☒ Example Triggers:

- 5 failed logins in 60 seconds ☒ Alert
- File integrity changes + new unknown process ☒ Incident
- Traffic to blacklisted IP ☒ Block via firewall integration

☒ 7. SIEM Benefits

- Centralized security visibility

- Real-time threat detection
- Compliance reporting (PCI-DSS, HIPAA)
- Faster incident response
- Threat intelligence integration